

```
In [1]: library(Seurat)
# library(SeuratDisk)

library(reticulate)
library(anndata)

library(ggplot2)
library(ggpubr)
library(pheatmap)
library(dplyr)
library(tidyr)
library(RColorBrewer)
library(clustree)
library(repr)
options(repr.plot.width=10, repr.plot.height=8)

library(UpSetR)
library(grid)

library(PRRoc)
library(Matrix)

getwd()

dataset_id <- "simulated_mm_RA"
genome_id <- "mm10"
level <- "subfamily"
samples <- c("young", "old", "all")

for (sample in samples) {
  dir.create(paste0("figures/figures_", dataset_id, "_subfamily_", sample))
}
dir.create(paste0("figures/figures_", dataset_id, "_subfamily"))

thrMinCells <- 500 * 0.05
```

Loading required package: SeuratObject

Loading required package: sp

'SeuratObject' was built under R 4.3.3 but the current version is 4.4.1; it is recommended that you reinstall 'SeuratObject' as the ABI for R may have changed

'SeuratObject' was built with package 'Matrix' 1.6.4 but the current version is 1.7.5; it is recommended that you reinstall 'SeuratObject' as the ABI for 'Matrix' may have changed

Attaching package: 'SeuratObject'

The following objects are masked from 'package:base':

intersect, t

Attaching package: 'anndata'

The following object is masked from 'package:SeuratObject':

Layers

Warning message:

"package 'dplyr' was built under R version 4.4.3"

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

Warning message:

"package 'tidyr' was built under R version 4.4.3"

Warning message:

"package 'RColorBrewer' was built under R version 4.4.3"

Loading required package: ggraph

Attaching package: 'ggraph'

The following object is masked from 'package:sp':

geometry

Loading required package: rlang

Warning message:

"package 'rlang' was built under R version 4.4.3"

Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

expand, pack, unpack

'/mnt/TEdata/TEbenchmarking/Rscripts'

```
Warning message in dir.create(paste0("figures/figures_", dataset_id, "_subfamily_", :  
  "'figures/figures_simulated_mm_RA_subfamily_young' already exists"  
Warning message in dir.create(paste0("figures/figures_", dataset_id, "_subfamily_", :  
  "'figures/figures_simulated_mm_RA_subfamily_old' already exists"  
Warning message in dir.create(paste0("figures/figures_", dataset_id, "_subfamily_", :  
  "'figures/figures_simulated_mm_RA_subfamily_all' already exists"  
Warning message in dir.create(paste0("figures/figures_", dataset_id, "_subfamily")):  
  "'figures/figures_simulated_mm_RA_subfamily' already exists"
```

```

In [ ]: ## SUBSTITUTE WITH YOUR OWN PATHS
soloTE_result_path <- "/mnt/volume_1p5T"
irescue_result_path <- "/mnt/TEdataStorage_2T/results"
scTE_result_path <- "/mnt/TEdataStorage_2T/results"
# path of annotation files downloaded by scTE
scTE_rmsk_txt <- "/mnt/TEdata/TEbenchmarking/data/TEannotation/scTE/rmsk.txt.gz"
scTE_gtf <- "/mnt/TEdata/TEbenchmarking/data/TEannotation/scTE/genome.vM21.annotation.gtf.gz"

In [2]: # function to clean symbols from string
alphanumeric <- function(s) {
  gsub("[^[:alnum:]]", "", s)
}

In [22]: # import workspace
load("workspaces/00_evaluation_objectCreation.Rdata")

In [ ]: colorTools <- c(
  "scTE" = "#E03756",
  "IRescue" = "#F88475",
  "STARsolo_TE" = "#FFBC81",
  "STARsolo_TE_EM" = "#F3E088",
  "SoloTE_subfamily" = "#CADE97",
  "SoloTE_unique" = "#A4DD9B",
  "SoloTE_thr2" = "#6FC69D",
  "SoloTE_thr1" = "#4BB2BB",
  "SoloTE_thr0" = "#3989BF",
  "Stellarscope" = "#4F5D93",
  "simulated" = "grey70"
)

In [23]: conversion_table$subfamily_commonNames <- gsub(pattern="-", replacement="_", conversion_table$subfamily)

```

Aggregate splatter matrices

```

In [24]: # Aggregate splatter matrices by subfamily
splatter_objs_subfamily <- list()

for(sample in samples){
  subfamilies <- conversion_table$subfamily[match(Features(splatter_objs[[sample]]), conversion_table$stella:

  print(table(conversion_table$class[match(Features(splatter_objs[[sample]]), conversion_table$stellarscopeID

  locus_counts <- GetAssayData(splatter_objs[[sample]], layer="counts")

  # Aggregate rows by subfamily using the 'by' function
  subfamily_counts <- by(locus_counts, subfamilies, colSums)
  # Convert the result to a proper matrix
  result_matrix <- do.call(rbind, subfamily_counts)

  splatter_objs_subfamily[[sample]] <- Seurat::CreateSeuratObject(result_matrix,
                                                                    project = splatter_objs[[sample
                                                                    min.cells = 0, min.features = 0

}
splatter_objs_subfamily

DNA LINE  LTR  RC SINE
1128  820 1693  93 1213

Warning message:
"Data is of class matrix. Coercing to dgCMatrx."
DNA LINE  LTR  RC SINE
1147  546  577  97  663

Warning message:
"Data is of class matrix. Coercing to dgCMatrx."
LINE  LTR SINE
286 1157  587

Warning message:
"Data is of class matrix. Coercing to dgCMatrx."

```

```

$all
An object of class Seurat
475 features across 500 samples within 1 assay
Active assay: RNA (475 features, 0 variable features)
  1 layer present: counts

$old
An object of class Seurat
447 features across 500 samples within 1 assay
Active assay: RNA (447 features, 0 variable features)
  1 layer present: counts

$young
An object of class Seurat
60 features across 500 samples within 1 assay
Active assay: RNA (60 features, 0 variable features)
  1 layer present: counts

```

Aggregate tool matrices

```

In [25]: objListSubfamily <- list()

for(tool in names(objList)){
  for(sample in samples){

    subfamilies <- conversion_table$subfamily[match(Features(objList[[tool]][[sample]]), conversion_table$stellatus)]

    locus_counts <- GetAssayData(objList[[tool]][[sample]], layer="counts")

    # Aggregate rows by subfamily using the 'by' function
    subfamily_counts <- by(locus_counts, subfamilies, colSums)
    # Convert the result to a proper matrix
    result_matrix <- do.call(rbind, subfamily_counts)

    objListSubfamily[[tool]][[sample]] <- Seurat::CreateSeuratObject(result_matrix,
                                                                    project = objList[[tool]][[sample]]$project,
                                                                    min.cells = thrMinCells, min.features = thrMinFeatures)
  }
}

names(objListSubfamily)

```



```

rownames(SoloTE_TEmat) <- gsub("SoloTE\\", "", rownames(SoloTE_TEmat))
rownames(SoloTE_TEmat) <- gsub("\\|", "-", rownames(SoloTE_TEmat))
rownames(SoloTE_TEmat) <- gsub("\\_", "-", rownames(SoloTE_TEmat))
rownames(SoloTE_TEmat) <- gsub("\\?", "", rownames(SoloTE_TEmat))

table(rownames(SoloTE_TEmat) %in% conversion_table$subfamily)

# create Seurat object with shallow filtering of TEs expressed in at least thrMinCells cells
SoloTE_subfamily_TEobjs[[sample]] <- Seurat::CreateSeuratObject(SoloTE_TEmat,
  project = paste0("SoloTE_subfamily_", sample),
  min.cells = thrMinCells, min.features = 0
)

print("Summary of n counts")
print(summary(SoloTE_subfamily_TEobjs[[sample]]$nCount_RNA))
print("Summary of n features")
print(summary(SoloTE_subfamily_TEobjs[[sample]]$nFeature_RNA))

print("Loci present in the annotation:")
print(table(rownames(SoloTE_subfamily_TEobjs[[sample]])) %in% conversion_table$stellarscopeID))

objListSubfamily[["SoloTE_subfamily"]][[sample]] <- SoloTE_subfamily_TEobjs[[sample]]

```

```

[1] "gene obj"
An object of class Seurat
208 features across 500 samples within 1 assay
Active assay: RNA (208 features, 0 variable features)
1 layer present: counts
TRUE
466
[1] "Summary of n counts"
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
41269  50763   53195   53193   55582   71612
[1] "Summary of n features"
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  92.0   103.0   105.0   105.3   108.0   118.0
[1] "Loci present in the annotation:"

TRUE
188

```

Import IRescue sub-family level matrix for young TEs

```

In [27]: sample <- "young"

main_path <- paste0(irescue_result_path, "/irescue/", dataset_id, "/", sample, "/counts/")

irescue_objs <- list()

# Read the three gzipped files
irescue_matrix <- Seurat::Read10X(main_path)

print("Col sums:")
summary(colSums(irescue_matrix))

# # change - to _ to find common subfamily names
rownames(irescue_matrix) <- gsub(pattern="-", replacement="_", rownames(irescue_matrix))
print("Subfamilies in conversion table:")
print(table(rownames(irescue_matrix) %in% conversion_table$subfamily_commonNames))

print("Total counts of the subfamilies not present in our annotation are all 0:")
rowSums(irescue_matrix[rownames(irescue_matrix) ! (rownames(irescue_matrix) %in% conversion_table$subfamily_comm

# remove rows with 0 counts
irescue_matrix <- irescue_matrix[rowSums(irescue_matrix)>0,]

# match to names used for other tools
rownames(irescue_matrix) <- conversion_table$subfamily[match(rownames(irescue_matrix), conversion_table$subfami

print("Non-zero subfamilies in conversion table:")
table(rownames(irescue_matrix) %in% conversion_table$subfamily)

irescue_objs[[sample]] <- Seurat::CreateSeuratObject(irescue_matrix,
  project = paste0("IRescue_", sample),
  min.cells = thrMinCells, min.features = 0
)
irescue_objs

objListSubfamily[["IRescue"]][[sample]] <- irescue_objs[[sample]]

```

```
[1] "Col sums:"
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
23439  37069  40927  41093  44896  65719
[1] "Subfamilies in conversion table:"

FALSE TRUE
      8  921
[1] "Total counts of the subfamilies not present in our annotation are all 0:"
LFSINE_Vert: 0 MamSINE1: 0 U1: 0 U17: 0 U2: 0 U5: 0 U6: 0 U7: 0

[1] "Non-zero subfamilies in conversion table:"
TRUE
212
$young
An object of class Seurat
95 features across 500 samples within 1 assay
Active assay: RNA (95 features, 0 variable features)
1 layer present: counts
```

Import scTE sub-family level matrix for young TEs

```
In [28]: sample <- "young"

main_path <- paste0(scTE_result_path, "/scTE/", sample, ".csv")

scTE_objs <- list()

# read matrix
scTE_matrix <- read.csv(main_path, row.names=1)
scTE_matrix <- as.matrix(scTE_matrix)
scTE_matrix <- t(scTE_matrix)

print("Col sums:")
summary(colSums(scTE_matrix))

# remove rows with 0 counts
scTE_matrix <- scTE_matrix[rowSums(scTE_matrix)>0,]

summary(rowSums(scTE_matrix))
```

```
[1] "Col sums:"
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
25564  38157  41618  41685  45116  63563
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
         1         1         6    6881        36 2677306
```

```
In [29]: # import annotations used by scTE
scTE_TEannotation <- read.table(scTE_rmsk_txt)
head(scTE_TEannotation)

scTE_gene_gtf <- read.table(scTE_gtf, sep="\t", comment.char="#", header=FALSE,
                           col.names=c("seqname","source","feature","start","end","score","strand","frame","attributes"))
scTE_gene_gtf$gene_name <- gsub('.*gene_name ([^;]+);.*', '\\1', scTE_gene_gtf$attributes)

scTE_genes <- unique(scTE_gene_gtf$gene_name)
head(scTE_genes)
```

A data.frame: 6 × 17

	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12	
	<int>	<int>	<int>	<int>	<int>	<chr>	<int>	<int>	<int>	<chr>	<chr>	<chr>	<
1	607	12955	105	9	10	chr1	3000000	3002128	-192469843	-	L1_Mus3	LINE	
2	607	1216	268	31	105	chr1	3003152	3003994	-192467977	-	L1Md_F	LINE	
3	607	234	279	0	0	chr1	3003993	3004054	-192467917	-	L1_Mus3	LINE	
4	607	3685	199	21	14	chr1	3004040	3004206	-192467765	+	L1_Rod	LINE	
5	607	376	62	31	0	chr1	3004206	3004270	-192467701	+	(CAAA)n	Simple_repeat	Simple_re
6	607	3685	199	21	14	chr1	3004270	3005001	-192466970	+	L1_Rod	LINE	

```
'4933401J01Rik' · 'Gm26206' · 'Xkr4' · 'Gm18956' · 'Gm37180' · 'Gm37363'
```

```
In [30]: # check if our simulated subfamilies are in the scTE TE annotation
table( alphanumeric(rownames(splatter_objs_subfamily[["young"]])) %in% alphanumeric(scTE_TEannotation$V11))

TRUE
60
```

```
In [31]: # create df of scTE features and annotate them as genes and TEs
```

```

scTE_feature_df <- as.data.frame(rownames(scTE_matrix))
colnames(scTE_feature_df) <- "scTE_name"

scTE_feature_df$feature_type <- "unknown"

scTE_feature_df$feature_type[alphanumeric(scTE_feature_df$scTE_name) %in% alphanumeric(scTE_TEannotation$V11)] <- "TE"
scTE_feature_df$feature_type[alphanumeric(scTE_feature_df$scTE_name) %in% alphanumeric(scTE_genes)] <- "gene"
scTE_feature_df$feature_type[grepl("Rik", scTE_feature_df$scTE_name)] <- "gene"

table(scTE_feature_df$feature_type)

```

```

gene    TE
2454    575

```

```

In [32]: # Check how many detected genes are expressed in more than 5% of cells
scTE_matrix_genes <- scTE_matrix[scTE_feature_df$scTE_name[scTE_feature_df$feature_type=="gene"],]
dim(scTE_matrix_genes[rowSums(scTE_matrix_genes) > thrMinCells,])

```

```
615 500
```

```

In [33]: # keep only TEs in the scTE matrix
scTE_matrix <- scTE_matrix[scTE_feature_df$scTE_name[scTE_feature_df$feature_type=="TE"],]
dim(scTE_matrix)

```

```
575 500
```

```

In [34]: # change - and . to _ to find common subfamily names
rownames(scTE_matrix) <- gsub(pattern="-", replacement="_", rownames(scTE_matrix))
rownames(scTE_matrix) <- gsub(pattern="\\.", replacement="_", rownames(scTE_matrix))

print("Subfamily not present in our annotation:")
rownames(scTE_matrix)[!(rownames(scTE_matrix) %in% conversion_table$subfamily_commonNames)] # not in our conversion table

# match to names used for other tools (for those that are matched)
matched <- conversion_table$subfamily[match(rownames(scTE_matrix), conversion_table$subfamily_commonNames)]
rownames(scTE_matrix) <- ifelse(is.na(matched), rownames(scTE_matrix), matched)

print("Non-zero subfamilies in conversion table:")
table(rownames(scTE_matrix) %in% conversion_table$subfamily)

scTE_objs[[sample]] <- Seurat::CreateSeuratObject(scTE_matrix,
  project = paste0("scTE_", sample),
  min.cells = thrMinCells, min.features = 0
)
scTE_objs

objListSubfamily[["scTE"]][[sample]] <- scTE_objs[[sample]]

```

```
[1] "Subfamily not present in our annotation:"
```

```
'LFSINE_Vert'
```

```
[1] "Non-zero subfamilies in conversion table:"
```

```
FALSE TRUE
1 574
```

Warning message:

"Feature names cannot have underscores ('_'), replacing with dashes ('-')"

Warning message:

"Data is of class matrix. Coercing to dgCMatix."

```
$young
```

```
An object of class Seurat
```

```
260 features across 500 samples within 1 assay
```

```
Active assay: RNA (260 features, 0 variable features)
```

```
1 layer present: counts
```

Statistics

```

In [35]: for(sample in samples){
  print(sample)
  print(splatter_objs_subfamily[[sample]]@project.name)
  print(sum(splatter_objs_subfamily[[sample]]$nCount_RNA))
  nCounts_sample <- vector()

  for(tool in names(objListSubfamily)){
    if(sample %in% names(objListSubfamily[[tool]])){
      obj <- objListSubfamily[[tool]][[sample]]
      nCounts_sample[obj@project.name] <- sum(obj$nCount_RNA)
    }
  }
  print(nCounts_sample)
}

```



```

[1] "all"
[1] "simulated_all"
[1] 30376911
      STARsolo_TE_all STARsolo_TE_EM_all SoloTE_unique_all SoloTE_thr2_all
      20864548      28093185      19365845      20256586
      SoloTE_thr1_all SoloTE_thr0_all Stellarscope_all
      21087351      28829577      26733978
[1] "old"
[1] "simulated_old"
[1] 30384744
      STARsolo_TE_old STARsolo_TE_EM_old SoloTE_unique_old SoloTE_thr2_old
      29869669      30342764      31605699      31841727
      SoloTE_thr1_old SoloTE_thr0_old Stellarscope_old
      31924535      32271772      31666805
[1] "young"
[1] "simulated_young"
[1] 30392717
      STARsolo_TE_young STARsolo_TE_EM_young SoloTE_unique_young
      9424570      25014811      3805194
      SoloTE_thr2_young SoloTE_thr1_young SoloTE_thr0_young
      6337946      8279167      25812726
      Stellarscope_young SoloTE_subfamily_young IRescue_young
      22135230      26596330      20545865
      scTE_young
      20415903

```

In [36]: ## n TPS

```

for(sample in samples){
  print(sample)
  print("Simulated")
  print(length(Features(splatter_objs_subfamily[[sample]])))

  for(tool in names(objListSubfamily)){
    if(sample %in% names(objListSubfamily[[tool]])){
      obj <- objListSubfamily[[tool]][[sample]]
      print(obj@project.name)
      print(length(intersect(Features(splatter_objs_subfamily[[sample]]), Features(obj))))
    }
  }
}

```

```

[1] "all"
[1] "Simulated"
[1] 475
[1] "STARsolo_TE_all"
[1] 467
[1] "STARsolo_TE_EM_all"
[1] 475
[1] "SoloTE_unique_all"
[1] 459
[1] "SoloTE_thr2_all"
[1] 462
[1] "SoloTE_thr1_all"
[1] 463
[1] "SoloTE_thr0_all"
[1] 474
[1] "Stellarscope_all"
[1] 471
[1] "old"
[1] "Simulated"
[1] 447
[1] "STARsolo_TE_old"
[1] 438
[1] "STARsolo_TE_EM_old"
[1] 447
[1] "SoloTE_unique_old"
[1] 431
[1] "SoloTE_thr2_old"
[1] 432
[1] "SoloTE_thr1_old"
[1] 435
[1] "SoloTE_thr0_old"
[1] 446
[1] "Stellarscope_old"
[1] 442
[1] "young"
[1] "Simulated"
[1] 60
[1] "STARsolo_TE_young"
[1] 58
[1] "STARsolo_TE_EM_young"
[1] 60
[1] "SoloTE_unique_young"
[1] 54
[1] "SoloTE_thr2_young"
[1] 55
[1] "SoloTE_thr1_young"
[1] 55
[1] "SoloTE_thr0_young"
[1] 60
[1] "Stellarscope_young"
[1] 60
[1] "SoloTE_subfamily_young"
[1] 60
[1] "IRescue_young"
[1] 60
[1] "scTE_young"
[1] 59

```

In [37]: `options(repr.plot.width=12, repr.plot.height=6)`

```

for (sample in samples){

  print(sample)

  detectedList <- list()
  TPList <- list()
  # get tools that were run for the current sample and reorder based on color
  tools_with_sample <- names(Filter(function(tool) sample %in% names(tool), objListSubfamily))
  tools_with_sample <- names(colorTools)[names(colorTools) %in% tools_with_sample]

  for (tool in c(tools_with_sample, "simulated")){

    print(tool)
    if (tool == "simulated") {
      obj <- splatter_objs_subfamily[[sample]]
    }else {
      obj <- objListSubfamily[[tool]][[sample]]
    }
    print(obj)
    detectedList[[sub("^(.*)_*$", "\\1", obj@project.name)]] <- Features(obj)
    TPList[[sub("^(.*)_*$", "\\1", obj@project.name)]] <- intersect(Features(splatter_objs_subfamily[[sample]]
  }

```

```

pdf(file=paste0("figures/figures_simulated_mm_RA_subfamily_",sample,"/upset_detected.pdf"), width = 13, hei
show(UpSetR::upset(fromList(detectedList), nintersects = 30,
  sets=c(tools_with_sample, "simulated"), keep.order = T, sets.bar.color=colorTools[c(tools_with_samp
  text.scale = c(2, 2, 2, 1.65, 2.5, 1.5),
  order.by=c("freq"),
  point.size=3, mb.ratio = c(0.5, 0.5),
  set_size.show=F, set_size.scale_max= max(lengths(detectedList))+10, #show.numbers = F,
  nsets=length(c(tools_with_sample, "simulated")),
  sets.x.label="N. detected loci",
  mainbar.y.label="Intersection size")
)
grid.text(paste0("Detected TE subfamilies - ", sample ),x = 0.65, y=0.95, gp=gpar(fontsize=18))
dev.off()

pdf(file=paste0("figures/figures_simulated_mm_RA_subfamily_",sample,"/upset_detected_TP.pdf"), width = 11, l
show(UpSetR::upset(fromList(TPlist), nintersects = 20,
  sets=c(tools_with_sample, "simulated"), keep.order = T, sets.bar.color=colorTools[c(tools_with_samp
  text.scale = c(2, 2, 2, 1.65, 2.2, 1.5),
  order.by=c("freq"),
  point.size=3, mb.ratio = c(0.5, 0.5),
  set_size.show=F, set_size.scale_max= max(lengths(TPlist))+10, #show.numbers = F,
  nsets=length(c(tools_with_sample, "simulated")),
  sets.x.label="N. correctly detected loci",
  mainbar.y.label="Intersection size")
)
grid.text(paste0("TP TE subfamilies- ", sample ),x = 0.65, y=0.95, gp=gpar(fontsize=18))
dev.off()
}

```

```

[1] "all"
[1] "STARsolo_TE"
An object of class Seurat
479 features across 500 samples within 1 assay
Active assay: RNA (479 features, 0 variable features)
  1 layer present: counts
[1] "STARsolo_TE_EM"
An object of class Seurat
496 features across 500 samples within 1 assay
Active assay: RNA (496 features, 0 variable features)
  1 layer present: counts
[1] "SoloTE_unique"
An object of class Seurat
479 features across 500 samples within 1 assay
Active assay: RNA (479 features, 0 variable features)
  1 layer present: counts
[1] "SoloTE_thr2"
An object of class Seurat
484 features across 500 samples within 1 assay
Active assay: RNA (484 features, 0 variable features)
  1 layer present: counts
[1] "SoloTE_thr1"
An object of class Seurat
489 features across 500 samples within 1 assay
Active assay: RNA (489 features, 0 variable features)
  1 layer present: counts
[1] "SoloTE_thr0"
An object of class Seurat
514 features across 500 samples within 1 assay
Active assay: RNA (514 features, 0 variable features)
  1 layer present: counts
[1] "Stellarscope"
An object of class Seurat
483 features across 500 samples within 1 assay
Active assay: RNA (483 features, 0 variable features)
  1 layer present: counts
[1] "simulated"
An object of class Seurat
475 features across 500 samples within 1 assay
Active assay: RNA (475 features, 0 variable features)
  1 layer present: counts

```

```
Warning message:
"`aes_string()` was deprecated in ggplot2 3.0.0.
i Please use tidy evaluation idioms with `aes()`.
i See also `vignette("ggplot2-in-packages")` for more information.
i The deprecated feature was likely used in the UpSetR package.
  Please report the issue to the authors."
Warning message:
"Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.
i The deprecated feature was likely used in the UpSetR package.
  Please report the issue to the authors."
Warning message:
"The `size` argument of `element_line()` is deprecated as of ggplot2 3.4.0.
i Please use the `linewidth` argument instead.
i The deprecated feature was likely used in the UpSetR package.
  Please report the issue to the authors."
```

```
[1] "old"
[1] "STARsolo_TE"
An object of class Seurat
438 features across 500 samples within 1 assay
Active assay: RNA (438 features, 0 variable features)
  1 layer present: counts
[1] "STARsolo_TE_EM"
An object of class Seurat
449 features across 500 samples within 1 assay
Active assay: RNA (449 features, 0 variable features)
  1 layer present: counts
[1] "SoloTE_unique"
An object of class Seurat
439 features across 500 samples within 1 assay
Active assay: RNA (439 features, 0 variable features)
  1 layer present: counts
[1] "SoloTE_thr2"
An object of class Seurat
440 features across 500 samples within 1 assay
Active assay: RNA (440 features, 0 variable features)
  1 layer present: counts
[1] "SoloTE_thr1"
An object of class Seurat
443 features across 500 samples within 1 assay
Active assay: RNA (443 features, 0 variable features)
  1 layer present: counts
[1] "SoloTE_thr0"
An object of class Seurat
460 features across 500 samples within 1 assay
Active assay: RNA (460 features, 0 variable features)
  1 layer present: counts
[1] "Stellarscope"
An object of class Seurat
442 features across 500 samples within 1 assay
Active assay: RNA (442 features, 0 variable features)
  1 layer present: counts
[1] "simulated"
An object of class Seurat
447 features across 500 samples within 1 assay
Active assay: RNA (447 features, 0 variable features)
  1 layer present: counts
[1] "young"
[1] "scTE"
An object of class Seurat
260 features across 500 samples within 1 assay
Active assay: RNA (260 features, 0 variable features)
  1 layer present: counts
[1] "IRescue"
An object of class Seurat
95 features across 500 samples within 1 assay
Active assay: RNA (95 features, 0 variable features)
  1 layer present: counts
[1] "STARsolo_TE"
An object of class Seurat
83 features across 500 samples within 1 assay
Active assay: RNA (83 features, 0 variable features)
  1 layer present: counts
[1] "STARsolo_TE_EM"
An object of class Seurat
105 features across 500 samples within 1 assay
Active assay: RNA (105 features, 0 variable features)
  1 layer present: counts
[1] "SoloTE_subfamily"
An object of class Seurat
188 features across 500 samples within 1 assay
Active assay: RNA (188 features, 0 variable features)
```

```

1 layer present: counts
[1] "SoloTE_unique"
An object of class Seurat
77 features across 500 samples within 1 assay
Active assay: RNA (77 features, 0 variable features)
1 layer present: counts
[1] "SoloTE_thr2"
An object of class Seurat
81 features across 500 samples within 1 assay
Active assay: RNA (81 features, 0 variable features)
1 layer present: counts
[1] "SoloTE_thr1"
An object of class Seurat
87 features across 500 samples within 1 assay
Active assay: RNA (87 features, 0 variable features)
1 layer present: counts
[1] "SoloTE_thr0"
An object of class Seurat
155 features across 500 samples within 1 assay
Active assay: RNA (155 features, 0 variable features)
1 layer present: counts
[1] "Stellarscope"
An object of class Seurat
92 features across 500 samples within 1 assay
Active assay: RNA (92 features, 0 variable features)
1 layer present: counts
[1] "simulated"
An object of class Seurat
60 features across 500 samples within 1 assay
Active assay: RNA (60 features, 0 variable features)
1 layer present: counts

```

```

In [39]: precisionTool <- list()
recallTool <- list()

for (sample in samples){
  layer <- "counts"

  matSplatter <- GetAssayData(splatter_objs_subfamily[[sample]], layer=layer)

  precisionTool[[sample]] <- list()
  recallTool[[sample]] <- list()

  tools_with_sample <- names(Filter(function(tool) sample %in% names(tool), objListSubfamily))

  for(tool in tools_with_sample){

    matTool <- GetAssayData(objListSubfamily[[tool]][[sample]], layer=layer)

    precisionTool[[sample]][[tool]] <- NULL
    recallTool[[sample]][[tool]] <- NULL

    for(cell in Cells(splatter_objs_subfamily[[sample]])){

      exprSplatter <- names(matSplatter[,cell])[matSplatter[,cell] > 0]
      # TEs expressed in the simulated matrix

      exprTool <- names(matTool[,cell])[matTool[,cell] > 0]
      # TEs detected by the tool

      TP <- intersect(exprSplatter, exprTool)
      nTP <- length(TP)

      FP <- setdiff(exprTool, exprSplatter)
      nFP <- length(FP)

      # add FP genes to the FP count
      if(tool %in% names(objGeneList)){
        FP_genes <- names(matTool_genes[,cell])[matTool_genes[,cell] > 0]
        nFP_genes <- length(FP_genes)
        nFP <- nFP + nFP_genes
      }

      FN <- setdiff(exprSplatter, exprTool)
      nFN <- length(FN)

      precision <- nTP / (nTP + nFP)
      precisionTool[[sample]][[tool]] <- c(precisionTool[[sample]][[tool]], precision)

      recall <- nTP / (nTP + nFN)
      recallTool[[sample]][[tool]] <- c(recallTool[[sample]][[tool]], recall)
    }
  }
}

```

```
}
```

```
In [ ]: options(repr.plot.width=5, repr.plot.height=3)

for (sample in samples){
  print(sample)

  precisionDf <- stack(precisionTool[[sample]])
  colnames(precisionDf) <- c("precision", "tool")

  recallDf <- stack(recallTool[[sample]])
  colnames(recallDf) <- c("recall", "tool")

  df <- merge(precisionDf, recallDf, by='tool')

  df$F1score <- (2 * df$precision * df$recall) / (df$precision + df$recall)

  df$tool <- factor(df$tool, levels=rev(names(colorTools)))

  # plot just the mean
  precisionDf <- stack(lapply(precisionTool[[sample]], mean))
  colnames(precisionDf) <- c("precision", "tool")

  recallDf <- stack(lapply(recallTool[[sample]], mean))
  colnames(recallDf) <- c("recall", "tool")

  Fscores <- sapply(names(precisionTool[[sample]]), function(tool){
    (2 * precisionTool[[sample]][[tool]] * recallTool[[sample]][[tool]]) /
    (precisionTool[[sample]][[tool]] + recallTool[[sample]][[tool]])
  })
  meanFscores <- colMeans(Fscores)
  FscoreDf <- stack(meanFscores)
  colnames(FscoreDf) <- c("F1score", "tool")

  df <- merge(precisionDf, recallDf, by=c("tool"))
  df <- merge(df, FscoreDf, by=c("tool"))

  df$tool <- factor(df$tool, levels=rev(names(colorTools)))
}
```

```
In [41]: meanF1score_age_df <- NULL

for(age in c("old", "young")){

  # plot just the mean value
  precisionDf <- stack(lapply(precisionTool[[age]], mean))
  colnames(precisionDf) <- c("precision", "tool")

  recallDf <- stack(lapply(recallTool[[age]], mean))
  colnames(recallDf) <- c("recall", "tool")

  Fscores <- sapply(names(precisionTool[[age]]), function(tool){
    (2 * precisionTool[[age]][[tool]] * recallTool[[age]][[tool]]) /
    (precisionTool[[age]][[tool]] + recallTool[[age]][[tool]])
  })

  meanFscores <- colMeans(Fscores)
  FscoreDf <- stack(meanFscores)
  colnames(FscoreDf) <- c("F1score", "tool")

  df <- merge(precisionDf, recallDf, by=c("tool"))
  df <- merge(df, FscoreDf, by=c("tool"))

  df$tool <- factor(df$tool, levels=rev(names(colorTools)))

  # create df for facet plot
  meanF1score_age_df <- rbind(meanF1score_age_df, cbind(df, age))
}
```

```
In [42]: options(repr.plot.width=8, repr.plot.height=4)

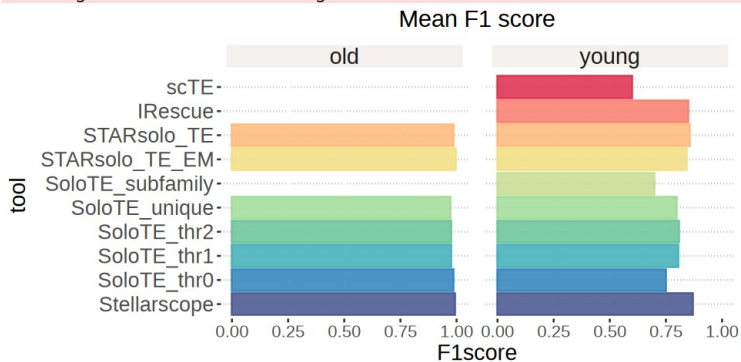
ggplot(meanF1score_age_df, aes(x=F1score, y=tool, color=tool, fill=tool)) +
  facet_wrap(~age, ncol = 2) +
  geom_col(alpha = 0.9) +
  scale_color_manual(values=colorTools) +
  scale_fill_manual(values=colorTools) +
  theme_pubclean() + ggtitle(paste0("Mean F1 score")) +
  guides(fill="none", color="none") +
  xlim(c(0,1)) +
  theme(text=element_text(size=17),
        plot.title = element_text(size=18, hjust=0.5),
```

```

plot.subtitle = element_text(size=18, hjust=0.5),
axis.text.y = element_text(size=16),
axis.text.x = element_text(size=13),
strip.text = element_text(size=17, hjust=0.5, vjust = 0.5),
strip.background = element_rect(fill="#F4F1EE", linewidth = 3, color = "white"),
panel.spacing = unit(1, "lines"))
ggsave(paste0("figures/figures_simulated_mm_RA_subfamily/meanF1_tool_byAge_horiz.png"))
ggsave(paste0("figures/figures_simulated_mm_RA_subfamily/meanF1_tool_byAge_horiz.pdf"), device="pdf", height=4,

```

Saving 6.67 x 6.67 in image



```

In [43]: options(repr.plot.width=9, repr.plot.height=6)
colorTools <- c(
  "scTE" = "#E03756",
  "IRescue" = "#F88475",
  "STARsolo_TE" = "#FFBC81",
  "STARsolo_TE_EM" = "#F3E088",
  "SoloTE_subfamily" = "#CADE97",
  "SoloTE_unique" = "#A4DD9B",
  "SoloTE_thr2" = "#6FC69D",
  "SoloTE_thr1" = "#4BB2BB",
  "SoloTE_thr0" = "#3989BF",
  "Stellarscope" = "#4F5D93",
  "simulated" = "grey70"
)
precision_plot <- ggplot(meanF1score_age_df, aes(x=precision, y=tool, color=tool, fill=tool)) +
  facet_wrap(~age, ncol = 1) +
  geom_col(alpha = 0.9) +
  scale_color_manual(values=colorTools) +
  scale_fill_manual(values=colorTools) +
  theme_pubclean() + ggtitle(paste0("Mean precision")) +
  guides(fill="none", color="none") +
  xlim(c(0,1)) +
  ylab("") +
  theme(text=element_text(size=17),
    plot.title = element_text(size=18, hjust=0.5),
    plot.subtitle = element_text(size=18, hjust=0.5),
    axis.text.y = element_text(size=16),
    axis.text.x = element_text(size=13),
    strip.text = element_text(size=17, hjust=0.5, vjust = 0.5),
    strip.background = element_rect(fill="#F4F1EE", linewidth = 3, color = "white"),
    panel.spacing = unit(1, "lines"))

recall_plot <- ggplot(meanF1score_age_df, aes(x=recall, y=tool, color=tool, fill=tool)) +
  facet_wrap(~age, ncol = 1) +
  geom_col(alpha = 0.9) +
  scale_color_manual(values=colorTools) +
  scale_fill_manual(values=colorTools) +
  theme_pubclean() + ggtitle(paste0("Mean recall")) +
  guides(fill="none", color="none") +
  xlim(c(0,1)) +
  ylab("") +
  theme(text=element_text(size=17),
    plot.title = element_text(size=18, hjust=0.5),
    plot.subtitle = element_text(size=18, hjust=0.5),
    axis.text.y = element_text(size=16),
    axis.text.x = element_text(size=13),
    strip.text = element_text(size=17, hjust=0.5, vjust = 0.5),
    strip.background = element_rect(fill="#F4F1EE", linewidth = 3, color = "white"),
    panel.spacing = unit(1, "lines"))

f1score_plot <- ggplot(meanF1score_age_df, aes(x=F1score, y=tool, color=tool, fill=tool)) +
  facet_wrap(~age, ncol = 1) +
  geom_col(alpha = 0.9) +
  scale_color_manual(values=colorTools) +
  scale_fill_manual(values=colorTools) +
  theme_pubclean() + ggtitle(paste0("Mean F1score")) +
  guides(fill="none", color="none") +
  xlim(c(0,1)) +

```

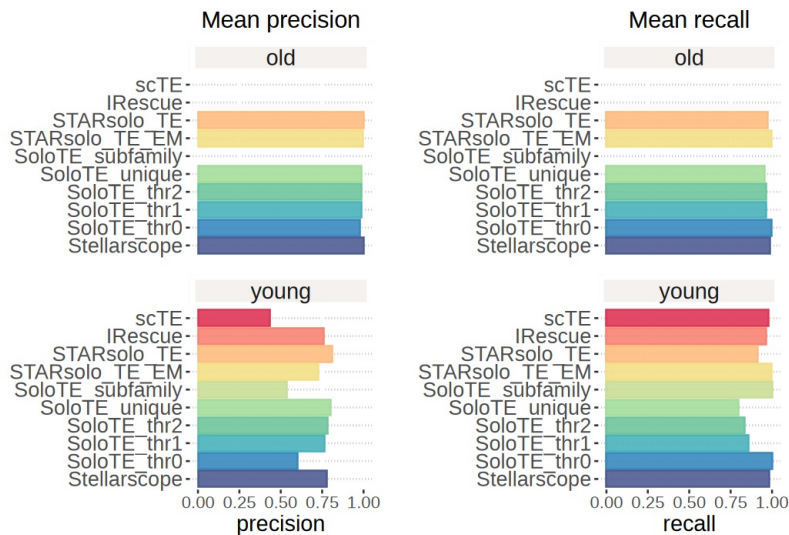
```

ylab("") +
theme(text=element_text(size=17),
plot.title = element_text(size=18, hjust=0.5),
plot.subtitle = element_text(size=18, hjust=0.5),
axis.text.y = element_text(size=16),
axis.text.x = element_text(size=13),
strip.text = element_text(size=17, hjust=0.5, vjust = 0.5),
strip.background = element_rect(fill="#F4F1EE", linewidth = 3, color = "white"),
panel.spacing = unit(1, "lines"))

precision_plot + recall_plot
ggsave(paste0("figures/figures_simulated_mm_RA_subfamily/meanPrecisionRecall_tool_byAge.png"),
height=6, width=9)

ggsave(paste0("figures/figures_simulated_mm_RA_subfamily/meanPrecisionRecall_tool_byAge.pdf"),
device="pdf", height=6, width=9)

```



```

In [44]: options(repr.plot.width=9, repr.plot.height=6)
colorTools <- c(
  "scTE" = "#E03756",
  "IRescue" = "#F88475",
  "STARsolo_TE" = "#FFBC81",
  "STARsolo_TE_EM" = "#F3E088",
  "SoloTE_subfamily" = "#CADE97",
  "SoloTE_unique" = "#A4DD9B",
  "SoloTE_thr2" = "#6FC69D",
  "SoloTE_thr1" = "#4BB2BB",
  "SoloTE_thr0" = "#3989BF",
  "Stellarscope" = "#4F5D93",
  "simulated" = "grey70"
)

precision_plot <- ggplot(meanF1score_age_df, aes(x=precision, y=tool, color=tool, fill=tool)) +
  facet_wrap(~age, ncol = 1) +
  geom_col(alpha = 0.9) +
  scale_color_manual(values=colorTools) +
  scale_fill_manual(values=colorTools) +
  theme_pubclean() + ggtitle(paste0("Mean precision")) +
  guides(fill="none", color="none") +
  xlim(c(0,1)) +
  ylab("") +
  theme(text=element_text(size=17),
    plot.title = element_text(size=18, hjust=0.5),
    plot.subtitle = element_text(size=18, hjust=0.5),
    axis.text.y = element_text(size=16),
    axis.text.x = element_text(size=13),
    strip.text = element_text(size=17, hjust=0.5, vjust = 0.5),
    strip.background = element_rect(fill="#F4F1EE", linewidth = 3, color = "white"),
    panel.spacing = unit(1, "lines"))

recall_plot <- ggplot(meanF1score_age_df, aes(x=recall, y=tool, color=tool, fill=tool)) +
  facet_wrap(~age, ncol = 1) +
  geom_col(alpha = 0.9) +
  scale_color_manual(values=colorTools) +
  scale_fill_manual(values=colorTools) +
  theme_pubclean() + ggtitle(paste0("Mean recall")) +
  guides(fill="none", color="none") +
  xlim(c(0,1)) +
  ylab("") +
  theme(text=element_text(size=17),
    plot.title = element_text(size=18, hjust=0.5),
    plot.subtitle = element_text(size=18, hjust=0.5),

```



```

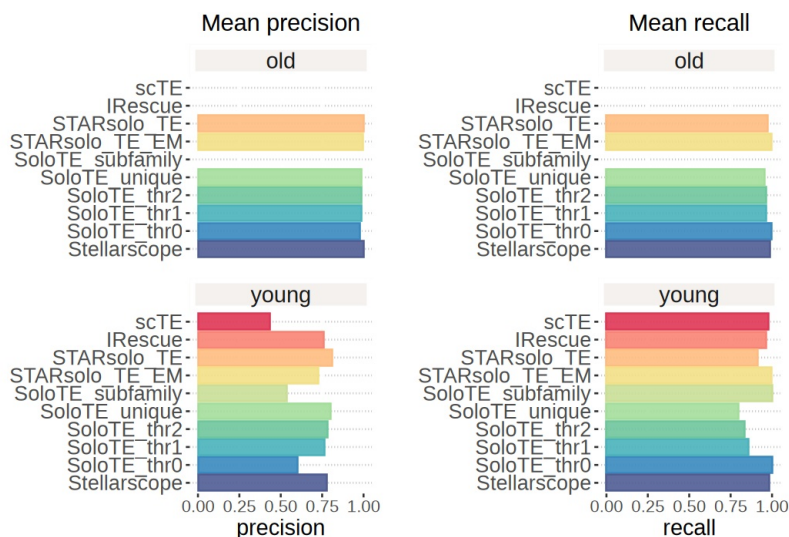
axis.text.y = element_text(size=16),
axis.text.x = element_text(size=13),
strip.text = element_text(size=17, hjust=0.5, vjust = 0.5),
strip.background = element_rect(fill="#F4F1EE", linewidth = 3, color = "white"),
panel.spacing = unit(1, "lines"))

f1score_plot <- ggplot(meanF1score_age_df, aes(x=F1score, y=tool, color=tool, fill=tool)) +
  facet_wrap(~age, ncol = 1) +
  geom_col(alpha = 0.9) +
  scale_color_manual(values=colorTools) +
  scale_fill_manual(values=colorTools) +
  theme_pubclean() + ggtitle(paste0("Mean F1score")) +
  guides(fill="none", color="none") +
  xlim(c(0,1)) +
  ylab("") +
  theme(text=element_text(size=17),
        plot.title = element_text(size=18, hjust=0.5),
        plot.subtitle = element_text(size=18, hjust=0.5),
        axis.text.y = element_text(size=16),
        axis.text.x = element_text(size=13),
        strip.text = element_text(size=17, hjust=0.5, vjust = 0.5),
        strip.background = element_rect(fill="#F4F1EE", linewidth = 3, color = "white"),
        panel.spacing = unit(1, "lines"))

precision_plot + recall_plot
ggsave(paste0("figures/figures_simulated_mm_RA_subfamily/meanPrecisionRecall_tool_byAge.png"),
       height=6, width=9)

ggsave(paste0("figures/figures_simulated_mm_RA_subfamily/meanPrecisionRecall_tool_byAge.pdf"),
       device="pdf", height=6, width=9)

```



```

In [45]: options(repr.plot.width=11, repr.plot.height=7)

library(patchwork)

recall_plot_noyname <- recall_plot + theme(axis.text.y = element_blank())
f1_plot_noyname <- f1score_plot + theme(axis.text.y = element_blank())

(precision_plot + recall_plot_noyname + f1_plot_noyname) +
  plot_layout(ncol = 3, guides = "collect") &
  theme(legend.position = "bottom")

ggsave(paste0("figures/figures_simulated_mm_RA_subfamily/meanPrecisionRecallF1score_tool_byAge.pdf"),
       device="pdf", width=11, height=7)

```

Warning message:
 "package 'patchwork' was built under R version 4.4.3"

